

TED (15)3072

MODEL QUESTION PAPER

**THIRD SEMESTER DIPLOMA EXAMINATION IN CHEMICAL
ENGINEERING**

FLUID MECHANICS

Time: 3hrs

Maximum marks: 100

PART –A

Maximum marks: 10

- I Answer the following questions in one or two sentences. Each question carries 2 marks.

- 1 State Newton's law of viscosity.
- 2 Define kinematic viscosity.
- 3 Define form friction.
- 4 Differentiate between simplex pump and duplex pump.
- 5 Define blower. (5X2=10)

PART –B

(Maximum marks:30)

- II Answer any five of the following questions. Each question carries 6 marks.

- 1 Name the pressure measuring devices commonly used in industry. 6
- 2 State the equation of continuity. 6
- 3 Explain the working of wet gas meter. 6
- 4 List the commonly used international pipe standards. 6
- 5 Draw the neat sketch of Globe valve and mark its parts. 6
- 6 Explain the working of diaphragm pump. 6
- 7 Explain the working of rotary blower. 6

(5X6=30)

PART – C

Maximum marks:60

(Answer one full question from each unit. Each full question carries 15 marks)

UNIT –I

- III (a) Water is flowing into a horizontal tapered pipe of diameter 400mm at the larger end and 200mm at the smaller end. The velocity of water at the larger end 5.5m/s. Compute the rate of discharge and velocity at the smaller end. 7
- (b) State and explain Bernoulli's theorem. 8

OR

- IV (a) Water is flowing through a pipe of 8cm of diameter under an absolute pressure of 5.5Kg/cm^2 and with mean velocity of 2.5m/s .Neglecting the friction losses, Calculate the total head. If the pipe is 7m above the datum line. 8
- (b) Explain the Reynolds experiment with a neat sketch. 7

UNIT II

- V (a) Explain the frictional losses in sudden contraction. 7
- (b) Brine of specific gravity 2.18 and viscosity $4.5 \times 10^{-3} \text{Ns/m}^2$ is to be pumped through 50m long smooth copper pipe having inside diameter 2.5cm. The flow rate 95l/minute. Estimate the pressure drop due to friction. How much power is required to overcome this friction? 8

OR

- VI (a) Explain the constructional details and working of Venturimeter. 8
- (b) Water is flowing at the rate of 2.5 Lit./s in a 20mm diameter pipe is metered by means of a simple orifice of diameter 15mm. If coefficient of discharge of orifice meter is 0.63. What will be the reading of the mercury manometer connected to the meter? 7

UNIT -III

- VII (a) Differentiate between pipe and tube. 7

	(b)	Explain the constructional details and working of stuffing box.	8
		OR	
VIII	(a)	Explain the commonly used pipe joining methods in chemical industry.	7
	(b)	Explain the constructional details and working of ball valve.	8
		UNIT-IV	
IX	(a)	Explain the constructional details and working Liquid piston type blower.	8
	(b)	Explain the constructional details and working of centrifugal pump.	7
		OR	
X	(a)	Explain the working of reciprocating compressor with sketch.	8
	(b)	Explain the working of reciprocating pump.	7